Robust VLAs through Cross-Modal Self-Prediction

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Predict missing inputs. Make robot policies more robust.

1 Motivation

78.6 → 87.5%

Zero-shot success on LIBERO-Plus

- Camera views, sensor noise and language change at deployment.

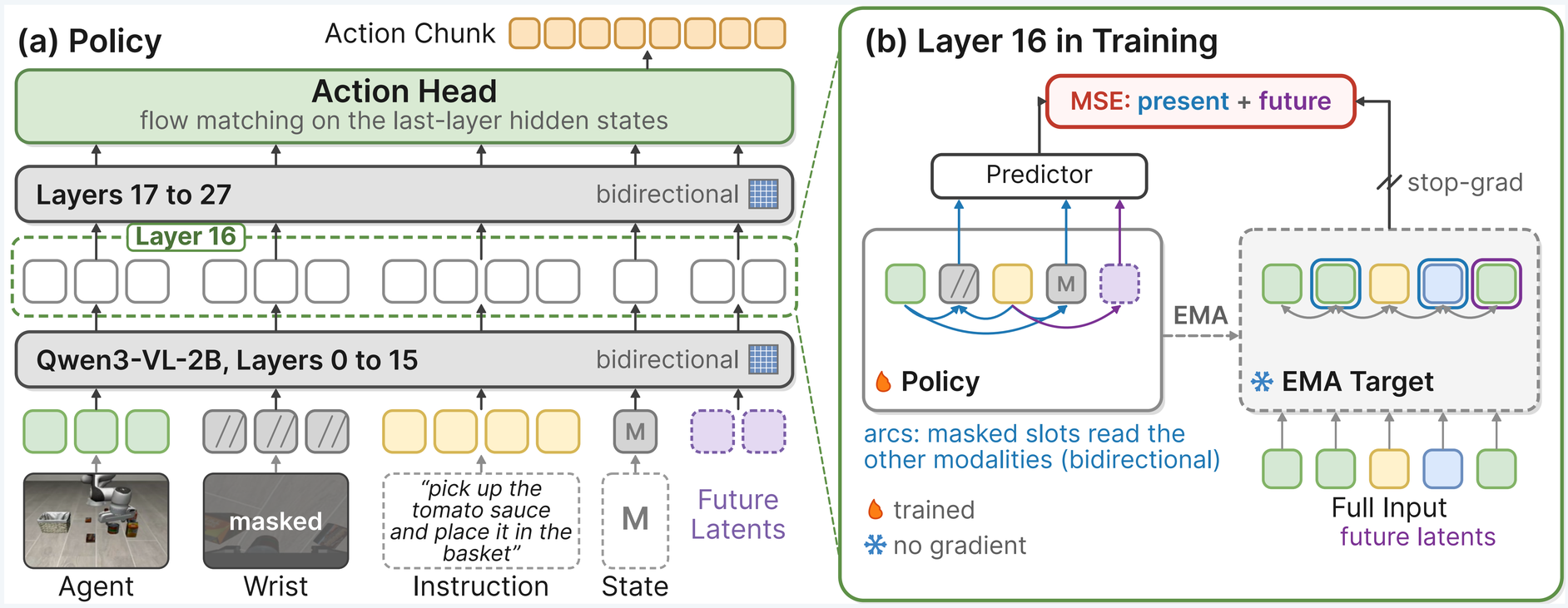
- **Can other inputs provide a fallback when one input changes?**



Key Idea

- Hide cameras, state and language during training.
- Predict full-input features with an EMA teacher.
- Deploy only the encoder and action head.

2 Method: Convert, then predict



Stage 2: action loss + 0.05 × prediction loss · full-input EMA teacher

Stage 1
Enc/dec conversion

Qwen3-VL-2B

↓

BiDir encoder

Causal decoder

6.72M embodied / spatial QA pairs

Causal PT: same data mixture

3 Results

Zero-shot LIBERO-Plus

87.5%

+8.9 points over encoder baseline

98.4%

Standard LIBERO success

98.2% with present-only CMSP

LIBERO ablations · success (%)

Plus: 3 seeds, mean ± s.d. · Clean: seed 42

Model / training	Plus	Clean
Causal PT baseline	77.9 ± 0.9	97.3
Causal PT + CMSP	83.1 ± 0.5	—
Encoder baseline	78.6 ± 1.1	96.3
+ Camera dropout	84.6 ± 0.4	97.2
+ All-modality masks	85.7 ± 0.3	97.4
+ CMSP prediction	87.0 ± 1.2	98.2
+ Future prediction	87.5 ± 0.7	98.4

Prediction adds +1.3 points over all-modality masking.

RoboCasa · camera loss

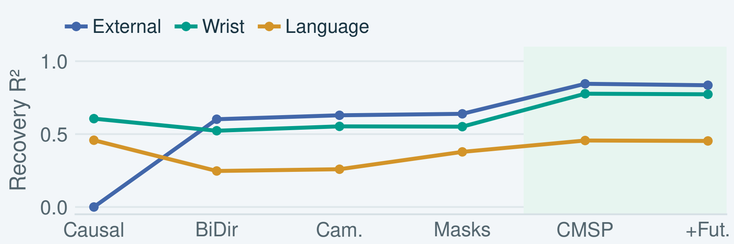
Matched 70k training; 18 × 50 episodes, seed 42

Training	Clean	No wrist
Camera dropout	81.7	45.2
All-modality masks	81.6	56.0
CMSP	79.4	56.0

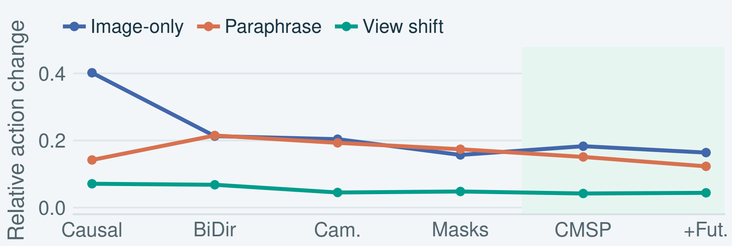
Masking alone matches CMSP with the wrist camera removed.

4 What does prediction change?

Missing-input recovery



Action sensitivity



Layer 16, seed 42 · recovery and sensitivity probes show associations, not causality.

Limitations

- Masking explains most of the robustness gain.

- RoboCasa: no prediction gain over masking.

- Unseen robot starts remain the weakest shift.

Karlsruhe Institute of Technology

CMSP

CoRL 2026 · World Models Workshop

Robotics Institute Germany

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